

Critical Minerals, Critical Pressures: Inflationary Dynamics in the Clean Energy Transition

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October 31, 2025

1 Motivation

The global shift toward decarbonization is transforming industrial demand for critical minerals such as lithium, nickel, copper, and cobalt. These materials are essential for batteries, electric vehicles, renewable energy infrastructure, and modern power grids. Unlike fossil fuels, their extraction and processing are geographically concentrated, investment-intensive, and environmentally constrained. As economies accelerate clean-energy investment, mineral bottlenecks have already produced sharp price surges—most notably, a sevenfold increase in lithium prices between 2020 and 2022.

Beyond resource scarcity, these constraints have important macroeconomic implications. The clean energy transition represents a large-scale reallocation shock that interacts with the business cycle, price stability, and welfare. Understanding how mineral scarcity affects inflation dynamics is therefore essential for designing coherent climate and monetary policy.

2 Conceptual Framework

The analysis develops a New Keynesian Climate (NKC) model in which clean-energy investment depends on critical minerals as intermediate inputs. Mineral scarcity acts as an endogenous supply-side constraint linking the cost of clean capital formation to global mineral prices. This mechanism generates a novel source of cost-push inflation distinct from traditional energy price shocks.

The framework highlights four key mechanisms:

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1. **Clean Investment Channel:** Clean capital accumulation requires mineral inputs whose supply elasticity is limited in the short run, making clean investment costs sensitive to mineral market tightness.
2. **Endogenous Cost-Push Inflation:** Tight mineral markets raise the marginal cost of clean investment, transmitting to aggregate inflation through intermediate input prices in the Phillips curve.
3. **Monetary-Climate Interaction:** The central bank faces a policy trade-off: stabilizing inflation through contractionary policy can inadvertently slow clean investment and delay decarbonization.
4. **Dynamic Feedback:** Learning-by-doing in clean technologies and endogenous mineral supply jointly determine cost declines, producing feedback between policy speed, inflation, and welfare.

This framework formalizes a green cost-push channel of inflation that complements existing analyses of redistribution, energy substitution, and carbon taxation.

3 Literature Review and Contribution

This project builds on emerging research at the intersection of climate macroeconomics, commodity markets, and monetary policy. Recent New Keynesian Climate models (e.g., Sahuc, Smets, and Vermandel (2025); Negro, Giovanni, and Dogra (2023)) highlight how decarbonization policies can generate short-run cost-push inflation through higher production costs and reallocation frictions. These frameworks reveal a policy trade-off between stabilizing inflation and sustaining investment during the clean transition, a tension that becomes more pronounced when critical inputs are scarce.

A parallel literature in resource and energy economics examines the role of critical minerals—such as lithium, nickel, and copper—in clean technology production. Studies by Boer, Pescatori, and Stuermer (2024) and Javorcik and Schweiger (2024) document that rapid growth in green demand, coupled with low short-run supply elasticity and geographic concentration of mining, can lead to prolonged price booms reminiscent of past oil shocks. These “green bottlenecks” may raise clean technology costs, slow decarbonization, and transmit upstream scarcity into aggregate inflation.

Empirical work by Miranda-Pinto et al. (2024) shows that metal supply shocks can have inflationary effects that are larger and more persistent than those of oil shocks, owing to metals’ central role as intermediate inputs across industries. Unlike oil shocks, which primarily

affect energy prices, metals shocks propagate through the production network and feed into core inflation, making them harder to overlook in monetary policy. Recent analyses by Avalos et al. (2025) argue that as the global economy becomes more metals-intensive, central banks will face new challenges in distinguishing transitory from persistent inflationary pressures.

This project extends these literatures by providing a unified macroeconomic framework that links mineral scarcity to inflation dynamics and welfare, clarifying the monetary-climate trade-off in the era of critical minerals.

4 Analytical Structure

The model features households, final-good producers, mineral suppliers, and policy authorities. Households maximize lifetime utility over consumption, labor, and environmental quality under nominal wage rigidity. Final-good firms combine clean and dirty energy through a CES aggregator, with clean energy requiring mineral-intensive capital. Mineral producers face convex extraction costs and sluggish capacity expansion, creating short-run scarcity rents. The monetary authority follows a Taylor rule targeting inflation, while the climate authority employs carbon pricing or investment subsidies to guide the energy mix.

The equilibrium links mineral supply constraints to inflation, output, and welfare. Faster decarbonization increases mineral demand and prices, raising production costs and triggering a contractionary monetary response. This interaction can slow clean investment despite long-run productivity and environmental gains, generating a temporary green stagflation. The framework thus formalizes a dynamic trade-off between transition speed and macroeconomic stability.

5 Contribution and Broader Relevance

By endogenizing mineral supply in a New Keynesian setting, the model provides a unified framework for studying macroeconomic frictions in the clean energy transition. It offers theoretical foundations for understanding the inflationary pressures observed during periods of rapid clean investment and suggests that effective coordination between monetary and climate policies is crucial to manage the transition.

The project’s broader contribution lies in clarifying how the pace of decarbonization must be matched by institutional and supply-chain adaptation to ensure that the transition remains both efficient and stable.

References

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Appendices

Appendix A Core Elements

The preliminary framework builds on a standard New Keynesian environment with nominal rigidities and environmental policy instruments. Aggregate output Y_t combines clean and dirty energy inputs:

$$Y_t = \left[\alpha (E_t^C)^{\frac{\sigma-1}{\sigma}} + (1-\alpha) (E_t^D)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

where σ is the elasticity of substitution between clean (E_t^C) and dirty (E_t^D) energy.

Clean energy production depends on installed clean capital K_t^C , which evolves as:

$$K_{t+1}^C = (1-\delta)K_t^C + \frac{I_t^C}{P_t^M},$$

where I_t^C denotes clean investment and P_t^M is the relative mineral price. Mineral scarcity enters through the supply curve

$$P_t^M = P_0^M \left(\frac{M_t}{M} \right)^\eta,$$

with $\eta > 0$ capturing the inverse elasticity of mineral supply.

Appendix B Inflation and Policy Interaction

Nominal rigidities follow a Calvo structure, producing the New Keynesian Phillips curve:

$$\pi_t = \beta E_t[\pi_{t+1}] + \kappa \tilde{m}_t,$$

where $\tilde{m}_t$ represents real marginal costs. Mineral scarcity affects inflation through input prices:

$$\tilde{m}_t = \phi_C \hat{P}_t^M + \phi_E \hat{P}_t^E.$$

An increase in mineral prices raises $\tilde{m}_t$, creating a direct cost-push channel.

The monetary authority follows

$$i_t = \rho_i i_{t-1} + (1-\rho_i)(\phi_\pi \pi_t + \phi_y y_t),$$

while the climate authority sets

$$\tau_t^C = \tau_0 + \chi(1 - s_t),$$

where s_t denotes the share of clean energy in total production. Accelerated decarbonization (higher s_t) increases mineral demand M_t , tightening monetary policy through the inflation channel and potentially dampening investment.

Appendix C Next Steps

The next stage of the project will formalize this system into a tractable dynamic stochastic model to examine transitional dynamics under alternative policy regimes. The analysis will explore how optimal monetary and climate policy coordination can balance inflation stabilization with the pace of decarbonization, providing a foundation for future empirical and quantitative extensions.